Quang Nguyen

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Education

Georgia Institute of Technology Atlanta, GA
M.S. in Electrical and Computer Engineering *Expected May 2028*

University of Rochester Rochester, NY
B.S. in Electrical and Computer Engineering, Minor in Computer Science, Cum Laude; GPA: 3.8/4.0 *May 2026*

Experience

Hardware Research Intern Rochester, NY
Laboratory for Laser Energetics *May 2025 – May 2026*

- Built an LTspice electrothermal model of various superconducting nanostrip single-photon detectors from device material properties and physical dimensions to simulate multi-photon electrical dynamics.
- Designed an RSFQ photon-number resolver in PSCAN2 using current-controlled DC-to-SFQ converters, splitters, and coincidence correlators (AND cells) that converts each detector waveform into a matched sequence of SFQ digital pulses.

Undergraduate Research Assistant Rochester, NY
University of Rochester *Feb 2025 – May 2025*

- Conducted pump-probe spectroscopy with femtosecond pulses to measure carrier relaxation in GaAs samples, producing response data sets for material characterization.
- Aligned the optics and synchronized the pulse trains for the strongest time-resolved signal in magneto-optic Kerr effect measurements with a rotating two-magnet field.

Hardware Engineer Intern Ho Chi Minh, Vietnam
Vita Imaging *May 2024 – July 2024*

- Designed a USB-UART serial board for a medical device, selecting an FT232RL and an LDO regulator for the 1.8/2.8 V rails, with ferrite beads and TVS diodes for ESD protection.
- Captured schematics in OrCAD and PCB layout in Allegro, applying high-speed design rules for trace width and differential-pair spacing. The prototype worked over USB 2.0.

Projects

Telepresence Rover | *WebRTC, Python, Node.js* May 2026

- Implemented WebRTC for a two-way audio and video link, connecting both peers on host candidates across a mesh VPN for long-range transmission, and pinning video to H.264 so the Pi encodes 720p at 24 fps in display hardware.
- Wrote the Python remote control, driving four motors at 1 kHz PWM and aiming a pan-tilt camera over I2C across the network, with a 0.35 s watchdog that halts the motors if the link drops.

MIPS 5-Stage Processor | *Verilog, Verilator* Dec 2025

- Designed a 32-bit MIPS processor in Verilog across 16 modules, with four pipeline registers carrying stall and flush control, an ALU, a branch unit, a register file, and a 530-line control unit decoding 57 integer instructions.
- Compiled the RTL to a C++ model with Verilator and debugged execution against compiled MIPS binaries from per-stage signal dumps.

Headphone Amplifier | *HSPICE, MATLAB* May 2025

- Designed a two-stage headphone amplifier with an OPA1656 preamp into a BUF634A output buffer, delivering 100 mW into 32 Ω with 0.72% THD for high-fidelity audio.
- Simulated stage gain, bandwidth, and input/output impedance in HSPICE. Soldered the board and compared measurements with waveform analysis in Python and MATLAB.

Technical Skills

Languages: Python, C/C++, MATLAB, Verilog, MIPS Assembly, Java.
Software: LTspice, HSPICE, PSCAN2, Verilator, Quartus, OrCAD, Allegro, KiCad, Git, Linux.
Hardware: Oscilloscope, Logic Analyzer, Multimeter, Waveform Generator, Soldering.
Coursework: VLSI Systems, Digital Logic Design, Analog IC, RFIC, Computer Architecture, Embedded Systems.